Brain Tumor MRI Classification Using Traditional Machine Learning

MSDIA S8 — Project 7

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Problem Statement

Motivation & goal

Brain tumor type strongly affects treatment decisions, but manual MRI interpretation is time-consuming and subject to variability. We target a reproducible, CPU-friendly pipeline suitable for decision support.

Research question: Can optimized handcrafted features + classical ML reach strong performance on a 3-class brain-tumor MRI task while remaining interpretable?

What we do (this talk)

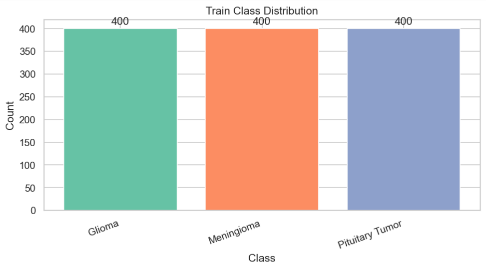
Step	Outcome
Feature extraction	Statistics, LBP, DWT, HOG descriptors
Feature optimization	Best params via separability metrics (no training)
Benchmarking	Tuned models across feature sets (8×8)
Validation	CV, bootstrap CI, McNemar / Friedman tests

Scope: no MRI theory deep dive; no algorithm derivations; focus on methodology and results.

Dataset

Dataset snapshot

Classes	3 (Glioma, Meningioma, Pituitary)
Total images	1 500 (500 per class)
Split	1 200 train / 300 test (80/20, strat
Image size	128 × 128 grayscale



Class distribution (train split).

Evaluation choice

We report macro-averaged metrics (especially macro-F1) so each tumor class contributes equally to the final score. With balanced classes, improvements reflect better separability rather than class reweighting.

Methodology Overview

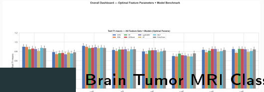
Dataset → Preprocessing → Feature Extraction → Feat… C… → … → …ation

Phase 1: Feature engineering

We search 213 descriptor configurations and select the best parameters using separability metrics (FDR, MI, DBI) *without* training classifiers.

Phase 2: ML benchmark

Freeze the optimized descriptors, then tune and compare 8 classifiers across 8 feature sets. Report held-out test performance + statistical validation.



Preprocessing

1) **Standardize** grayscale, rescale to $[0, 1]$

2) **Crop borders** remove black background

3) **Resize** 128×128 with anti-aliasing

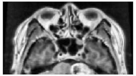
4) **Denoise** Gaussian smoothing ($\sigma = 0.5$)

5) **Enhance** min–max + CLAHE (local contrast)

Key setting	**Value**
Target size	128×128
Smoothing	$\sigma = 0.5$
Contrast	CLAHE (local)

Why this matters

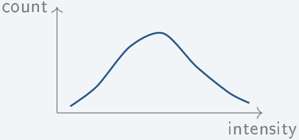
Preprocessing reduces acquisition variability and stabilizes the downstream descriptors (LBP texture, HOG gradients, DWT frequency).



Example preprocessed slice (CLAHE, 128×128).

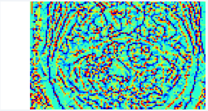
Feature Extraction Methods

A) Statistical features (intensity)



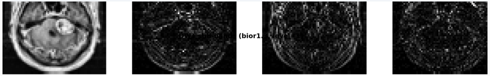
Compact summary of intensity distribution: mean, std, entropy, skew/kurtosis, and percentiles (p10–p90).

B) LBP (local texture)



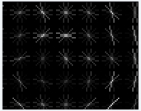
Encodes local micro-patterns; robust texture fingerprint for tumor tissue.

C) DWT (frequency sub-bands)



Multi-scale decomposition (LL/LH/HL/HH) captures coarse vs. fine structures.

D) HOG (edges & shape)



Aggregates gradient orientations; strong for boundary/shape cues.

Four complementary views of the same slice: intensity, texture, frequency, and gradients.

Optimization Strategy

Key idea

We avoid training classifiers for all 213 configurations.

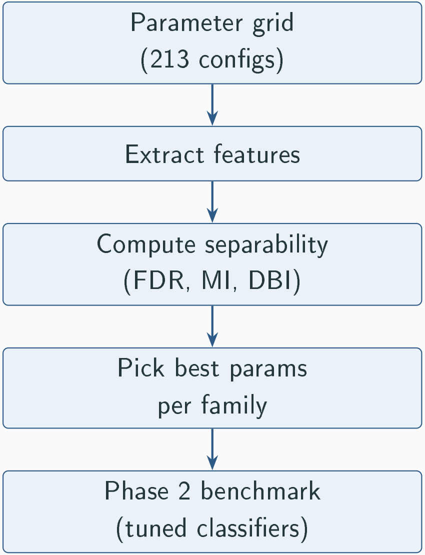
Instead, we select descriptor parameters using a **separability score** computed directly from the feature vectors:

$$\text{Score} = \frac{1}{3}\left(\widehat{\text{FDR}} + \widehat{\text{MI}} + (1 - \widehat{\text{DBI}})\right)$$

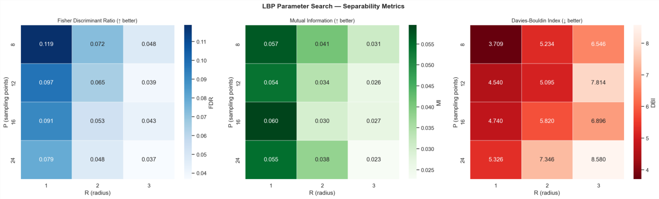
where hats denote min–max normalization across the search grid.

Why it works

High separability indicates class-specific structure before choosing a classifier, reducing the Phase 2



LBP Optimization Results



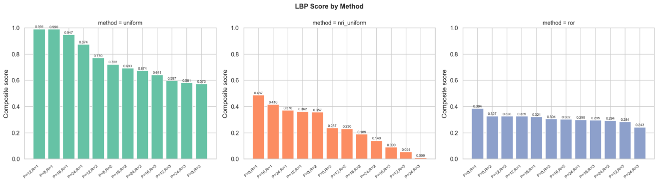
Selected configuration

Neighbors (P)	12
Radius (R)	1
Encoding	`uniform`
Separability	0.991

Takeaway

A small radius captures fine-grained micro-texture around tumor boundaries. Uniform encoding generalizes better by compressing patterns into a compact histogram.

LBP Method Comparison



Methods compared

Encoding	Property
`uniform`	compact + robust
`nri_uniform`	less invariant
`ror`	rotation-invariant

Decision

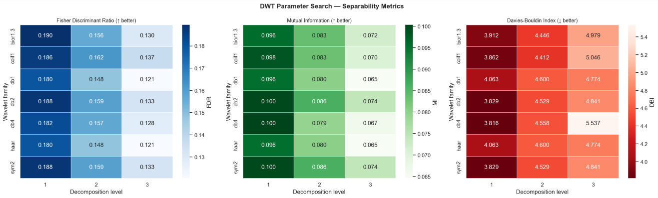
`uniform` wins consistently across (P, R) settings.

Selected for Phase 2 branch B.

Takeaway

Compact, rotation-tolerant patterns generalize best under

DWT Optimization Results



Search space

Wavelets	7 families
Levels	1–3
Total	21 configs

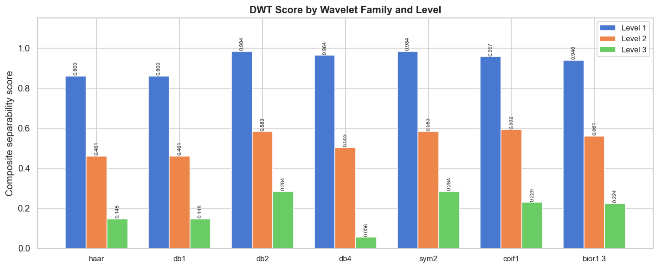
Selected configuration

Wavelet	db2
Level	1
Separability	0.984

Takeaway

db2 at level 1 captures discriminative mid-frequency structure without

DWT Wavelet Comparison



How to read this

We compare wavelet families under the same evaluation protocol.

Goal: maximize separability while keeping representation compact.

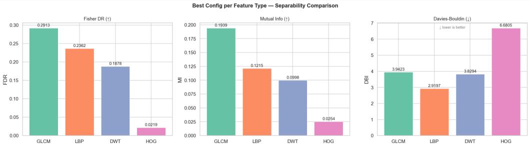
Winner

`db2` provides the strongest overall separability.

Interpretation: medium-frequency bands capture tumor tissue heterogeneity better than very coarse (haar) or overly complex families.

Phase 1 Summary

Selected parameters (frozen for Phase 2)



Feature	Best parameters	Separability
Statistics	11 first-order intensity features	—
LBP	P=12, R=1, uniform	0.991
DWT	`db2`, level $= 1$	0.984
HOG	6 orient.; cell $(16, 16)$; block $(3, 3)$; L2-Hys	0.990

Key observation

Texture (LBP) and gradients (HOG) achieve the highest separability.

This motivates using them as strong baselines and testing whether fusion adds value.

Feature Sets Evaluated

Building blocks (single families)

ID	Name	Signal captured
A	Statistics	intensity distribution (11 features)
B	LBP (opt)	local texture micro-patterns
C	DWT (opt)	multi-scale frequency sub-bands
D	HOG	edge/shape gradients

Fusion sets (incremental)

Set	Purpose
A+B	add texture to intensity
B+C(opt)	combine texture + frequency

Single families
A, B, C, D

Pair fusion
A+B, B+C

Triple fusion
A+B+C

Full(opt)
A+B+C+D

Rationale

Progressive fusion reveals which descriptors add complementary value beyond HOG.

Classifiers Compared

Model families (intuition)

Family	Models in benchmark
Margin / kernel	SVM (captures non-linear boundaries)
Instance-based	KNN (local neighborhoods)
Tree ensembles	Random Forest, ExtraTrees (bagging)
Boosted trees	XGBoost, LightGBM (strong tabular baselines)
Linear baseline	Logistic Regression
Shallow NN	MLP

Feature set (8)

↓

GridSearchCV (5-fold)

↓

Held-out test

8 models $\times$ 8 sets = **64** tuned configurations.

Evaluation Metrics

What we measure (macro-averaged)

Metric	Why it matters
Accuracy	overall correctness
Precision	avoid false positives (per class)
Recall	avoid missed tumors (per class)
F1-macro	balances precision/recall equally across classes
κ	agreement beyond chance (robustness)

Ranking criterion

We primarily rank configurations by **macro-F1** to ensure fair performance across the three tumor classes.

Train split (80%)

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5-fold CV + tuning

↓

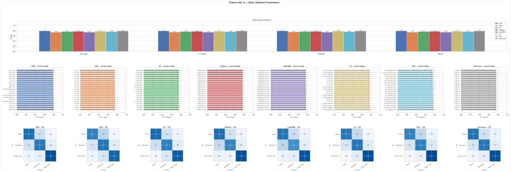
Held-out test (20%)

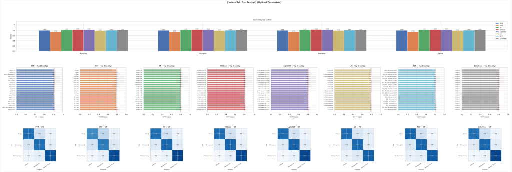
↓

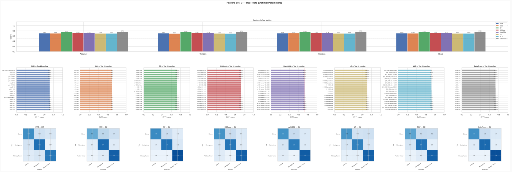
CI + significance tests

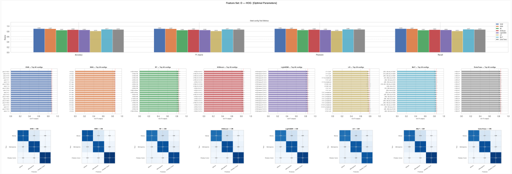
Bootstrap CI, McNemar (paired test errors), Friedman (multi-model comparison).

Individual Feature Results



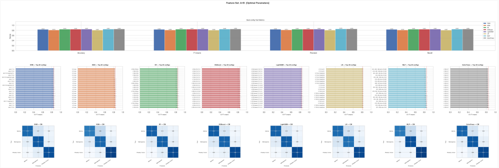


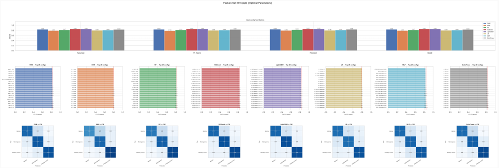


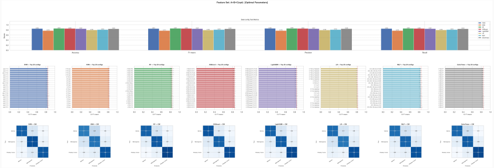


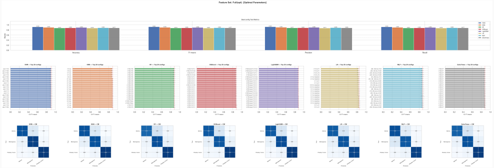
Best per branch: A/SVM 78.7% B/XGB 82.7% C/RF 75.7% D/SVM 90.3%

Combined Feature Results









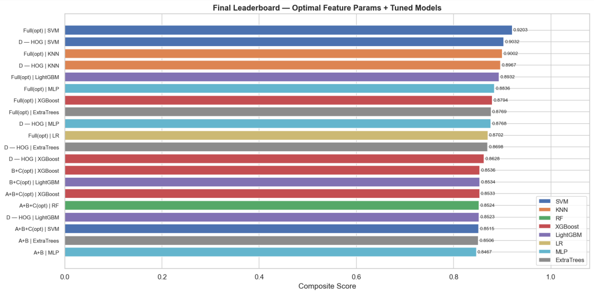
Fusion improves performance: Full(opt) reaches 92.0% (+ 1.7 pp over HOG alone).

Global Ranking

Top 5 (held-out test)

Rank	Feature set / Model	F1-macro
1	**Full(opt) / SVM**	**0.9204**
2	D — HOG / SVM	0.9032
3	Full(opt) / KNN	0.9002
4	D — HOG / KNN	0.8967
5	Full(opt) / LightGBM	0.8938

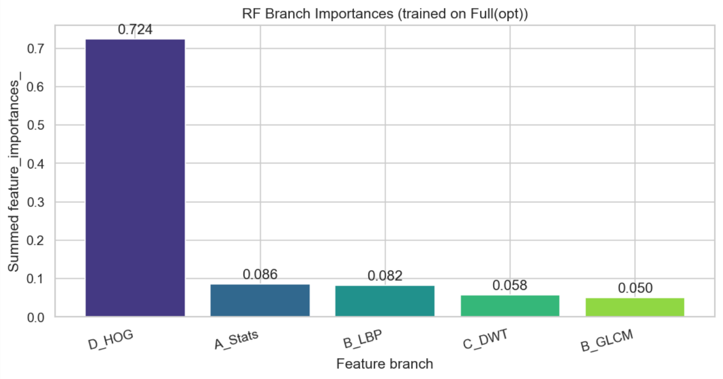
Table 1: Top 5 configurations (held-out test, ranked by F1-macro).



Key insight

Full(opt) + SVM ranks first (macro F1 0.9204)

Feature Importance Analysis



What this shows

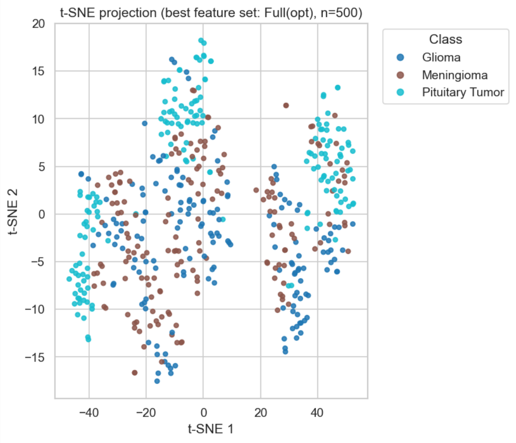
We group Full(opt) features by descriptor family and estimate their relative influence using a Random Forest surrogate model.

Takeaway

HOG and LBP dominate, while Stats and DWT provide smaller complementary gains.

This matches the observed improvement from HOG →

Feature Space Visualization



What is plotted

2D t-SNE projection of 526-dimensional Full(opt) feature vectors.

Interpretation

Partial separation indicates discriminative power.

Glioma vs. meningioma overlap helps explain residual confusion even in the best model.

Stability Analysis

Seed stability (Full(opt) + SVM)

Seed	Accuracy	F1-macro
42	0.9500	0.9498
43	0.9100	0.9099
44	0.9133	0.9130
Mean ± CI	—	**0.9242 ± 0.0251**

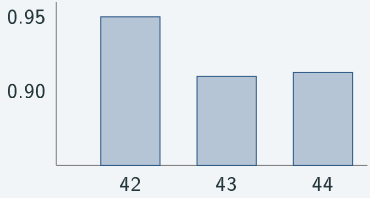
Table 2: Seed stability for Full(opt) + SVM (re-split per seed).

Setup

Three independent train/test splits (seeds 42, 43, 44).

Hyperparameters are re-tuned per seed (same protocol, different partition).

Quick visual (F1-macro)



Conclusion

Statistical Validation

Cross-validation (5-fold)

F1-macro	0.8951 ± 0.0100
Accuracy	0.8953 ± 0.0099

Stable performance across folds supports generalization.

Bootstrap CI (95%, test)

F1-macro	[0.8895, 0.9504]
Accuracy	[0.8900, 0.9500]

Uncertainty bounds on the held-out test estimate.

Significance tests

Test	**Result**
McNemar (SVM vs RF)	p=0.0056
McNemar (SVM vs XGB)	p=0.019
McNemar (SVM vs KNN)	p=0.39
Friedman (within Full)	p=0.0013

Confirms SVM improvements are meaningful vs several baselines.

Best Model

Final pipeline

Full(opt) + SVM

Full(opt) features
(526 cleaned)

↓

SVM (poly kernel)

↓

3-class prediction

Performance (held-out test)

Metric	Value
Accuracy	**92.0%**
Precision (macro)	0.9228
Recall (macro)	0.9200
F1-macro	**0.9204**
Cohen's κ	0.880

Uncertainty

95% bootstrap CI (macro-F1):
[0.8895, 0.9504]

Discussion

Why optimization + fusion help

Phase 1 ranks descriptor parameters using separability metrics, reducing trial-and-error in Phase 2.

Fusion combines complementary cues: intensity (A), texture (B), frequency (C), and edges (D).

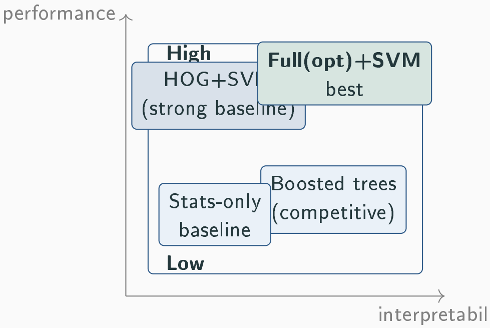
Why SVM performs best here

The polynomial kernel captures non-linear interactions in the 526-D Full(opt) space.

High-dimensional HOG features often favor margin-based learning over bagging-style ensembles.

Practical implications

CPU-only, reproducible, and descriptor-based



Trade-off sketch (relative positioning).

Conclusion and Future Work

Main contributions

Contribution	Outcome
Feature optimization	separability-driven parameter selection
Benchmarking	8 models $\times$ 8 feature sets
Validation	CV + bootstrap CI + significance tests
Best pipeline	**Full(opt)** + **SVM** at **92.0%**

Final takeaway

Optimized handcrafted features + classical ML can be strong on modest MRI datasets while remaining transparent and CPU-friendly.

Future work roadmap

CNNs / Vision Transformers

↓

Hybrid ML/DL
(handcrafted + learned)

↓

Explainable AI
(Grad-CAM, SHAP)

↓

External validation
(multi-center)